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Motor selection via impedance-matching for driving nonlinearly damped, resonant loads

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ABSTRACT

This paper presents a novel method for selecting the most appropriate motor–gearhead combination for nonlinearly damped, resonant loads. The method extends to the nonlinear damping case the impedance-matching condition which is used to guarantee a maximum power transfer in linear network theory. In particular, the method is applicable in general to resonant loads where the damping is an odd and memoryless nonlinear function of the velocity. This condition is very common in biomimetic robotics, in particular when designing propulsion systems based on flapping appendages, such as wings or fins, in viscous fluids, such as air or water. The method is graphical in nature and is based on a power vs. impedance-mismatch factor. Such a factor is function of the ratio of the motor armature resistance to the load equivalent resistive impedance reflected at the motor armature, where the latter is nonlinear and depends on the desired kinematics as well as on the transmission ratio and efficiency. The method allows comparing, for a given desired appendage kinematics, all motor–gearhead combinations at once while taking into account all possible constraints such as maximum current, power, and torque.

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1. Introduction

Electric motors are pervasive and can be found in nearly all modern machines. As the appropriate choice of the electric motor and the relative mechanical transmission directly affects the performance of the whole system, various methods have been presented to guide the Engineer during the motor-transmission selection [26,27,9,21,8]. Most industrial tasks require a motor to routinely perform work against typically inertial and resistive loads. Although this is definitely the most common loading condition for motors, other forms of loading are also possible which call for different selection methods. One example is the nonlinear, viscous damping generated by flapping fins and wings against water and air.

Recent years have witnessed an increase of research efforts in what is generally referred to as biomimetic robotics. Attracted by the unmatched performance of living systems, roboticists have started applying design principles drawing inspiration from biological evidence. In particular, agility and maneuverability in water or in air of living swimmers and flyers has inspired the development of an increasing number of so-called Micro Underwater Vehicles (MUVs) and Micro Aerial Vehicles (MAVs) [1,14,13]. Besides bio-inspired sensing capabilities and neuro-inspired forms of controllers [15], there has been a technological push towards the development of biomimetic forms of propulsion, with particular emphasis on flapping fins and wings. Both in air and in water, flapping locomotion is superior to other forms of propulsions especially at lower speeds. Unparalleled by man-made vehicles, animals such birds, bats, insects and also negatively buoyant fishes are in fact capable of fast forward motion as well as hovering, which is one of the most energetically challenging forms of locomotion since it cannot exploit the accumulated kinetic energy of the body as in forward swimming/flying [28,3].

Efficient power usage is fundamental for the development of flapping propellers. One of the limits to flapping propulsion, also faced by living systems especially at larger size, is represented by the inertia of the wings/fins. The need to periodically accelerate/decelerate the inertia of the appendices poses serious constraints to the flapping modality. Although the primary interest is doing work against the fluid, as this directly translates into production of lift and thrust forces, it is not uncommon that accelerating/decelerating wings/fins at relatively high frequencies might require much larger inertial torques than damping ones. This would lead to oversized muscles (and actuators for artificial systems).

A first biological observation concerns the fundamental role played by *resonance*, physiologically induced by the structural elasticity typically inherent to the biomechanical properties of muscles and, for insects, of the external cuticle. The presence of elastic components for storage and release of the kinetic energy of the wings/fins avoids the unnecessary waste of inertial power [28]. Indeed, such a principle has been successfully adopted in the design of flapping wings and fins for robotic applications [30,6,2].

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A second biological observation, which is key to the proposed selection method, is that the wings/fins kinematics observed in nature can, with a very good approximation, be considered *harmonic*. In particular, throughout this paper, this will be referred to as the *sinusoidal regime assumption*.

Based on the above considerations, this paper presents a novel method for selecting the appropriate motor–gearhead combination in order to directly drive a resonant mechanical system characterized by a typically nonlinear aero/fluid-dynamic damping. The nature of the nonlinearity is better detailed in the next section and, although the calculations are based on this particular form of nonlinear damping, the method is applicable whenever the nonlinearity is *memoryless* and *odd*. The proposed method is based on the matching impedance principle, a well known condition for linear networks which allows the maximum transfer of power from a fixed-voltage source to a load. The method is extended to the case of nonlinear resistive loads exploiting the sinusoidal regime condition and analytical techniques which are usually referred to 'linear approximant' or 'describing function' methods [11,22,23].

Finally, numerical tests are also presented in support of the sinusoidal regime assumption.

2. Fluid dynamical nonlinear damping

From a fluid dynamical perspective, the dynamics of hovering are qualitatively similar in air as much as underwater and insects, birds, bats and negatively buoyant fishes can be treat as a single functional group [3].

When a fin or a wing is moved relative to the surrounding fluid, energy is transferred to the fluid and reaction forces arise. In principle, the force distribution on the fin/wing as well as flow field of the surrounding fluid may be derived from the Navier–Stokes equations. In practice, numerical approaches (computational fluid dynamics) are the only viable ones for accurate solutions. Nevertheless, classically, reliable simplifications are based on the assumption of steady or quasi-steady flow [5,18]. Such models are based on quasi-steady blade element analysis where the wing/fin is assumed to be divided into a finite number of strips and each is analyzed independently.

For a wing/fin of length R, at distance r from the *fulcrum*, consider sections of infinitesimal area $c(r)dr$, where $c(r)$ is the wing chord which determines the geometric profile. For each section, the *instantaneous* drag[1] force is

$$dD = \frac{1}{2}\rho C_D (r\omega)^2 \mathrm{sign}(\omega) c(r) dr$$

where ρ is the density of the fluid (for air $\rho_{air} = 1.2$ kg/m^3 and for water $\rho_{water} = 10^3$ kg/m^3), C_D is the adimensional drag coefficient, and $r\omega$ is the linear velocity of the section. C_D depends on the angle of attack (i.e. inclination of the wing/fin with respect to the fluid velocity), therefore is in principle time dependent and can be averaged out throughout the motion [29]. C_D is close to zero for small angles of attack and monotonically increases up to a maximum value as the angle of attack approaches $\pi/2$. Maximal values range from 3.5 for some insects [10], to 0.4 for hummingbirds [28], to 2 for some fishes [3].

On any section, a drag torque $dB = rdD$ is induced by the drag force, i.e.

$$dB = \frac{1}{2}\rho C_D r (r\omega)^2 \mathrm{sign}(\omega) c(r) dr$$

[1] For underwater swimming animals and vehicles, drag refers to the whole body effect while thrust is considered for the fin only, therefore they are related by the body-area to foil-area ratio [16]. In this paper, only the fluid dynamical effects on the wing or fin are considered and drag shall be equivalent to thrust.

For a given wing profile $c(r)$, the above equations can be integrated over the whole wing length, leading to:

$$B(\omega) = \int_0^R dB = B_0 \omega^2 \mathrm{sign}(\omega) \tag{1}$$

with the torque damping coefficient B_0 defined as:

$$B_0 = \frac{1}{2}\rho C_D \int_0^R r^3 c(r)\, dr$$

where the integral depends on the particular wing/fin profile and its analytical evaluation is given in [29] for various wing shapes. For simulation purposes, numerical values for parameters were derived from [12] in the case of a box-fish and are summarized in Table 1.

The nonlinear damping $B(\omega)$ in (1) is therefore *memoryless*, since does not depends explicitly on time, and *odd*, meaning that $B(-\omega) = -B(\omega)$.

3. Sinusoidal regime assumption

Biological observations [29,17,25] provide evidence that: (i) mechanical resonance is the mechanism through which animals efficiently flap appendages; (ii) the resulting kinematics is with good approximation *sinusoidal*. Inherent elasticity in the musculo-skeletal structure of animals plays a key role in storing elastic strain energy which, especially in cyclic movements such as flapping appendages, can be released to efficiently enhance the force, work or power output.

Based on such observations, the *sinusoidal regime assumption* will allow estimating power requirements at steady-state: for a given *stroke-angle* $\pm\theta_0$ and a given flapping frequency f_0, the angular wing motion (θ), angular wing velocity ($\omega = \dot{\theta}$) and the angular wing acceleration ($\alpha = \ddot{\theta}$) in *sinusoidal regime* (denoted by the $\sim$ symbol) can be expressed as

$$\tilde{\theta} = \theta_0 \sin(2\pi f_0 t) \tag{2}$$

$$\tilde{\omega} = \Omega_0 \cos(2\pi f_0 t) \tag{3}$$

$$\tilde{\alpha} = -2\pi f_0 \Omega_0 \sin(2\pi f_0 t) \tag{4}$$

where the for the angular speed amplitude Ω_0, the following definition will be conveniently adopted:

$$\Omega_0 \stackrel{def}{=} 2\pi f_0 \theta_0 \tag{5}$$

At sinusoidal regime, the nonlinear damping torque $B(\tilde{\omega})$ defined in (1), can be expressed in terms of an equivalent proportional damping $B_1\tilde{\omega}$, where B_1 is defined to provide the same average power (over a period $T = 1/f_0$):

$$\frac{1}{T}\int_T B_1 \tilde{\omega}^2\, dt = \frac{1}{T}\int_T B(\tilde{\omega})\tilde{\omega}\, dt$$

i.e.

$$\int_T B_1 \tilde{\omega}^2\, dt = \int_T B_0 \tilde{\omega}^3 \mathrm{sign}(\tilde{\omega})\, dt$$

With change of variable ($2\pi f_0 t \to x$), this can conveniently computed over a quarter of period

Table 1
Parameters used for simulation. Numerical values were derived from [12].

Description	Parameter	Value	Units
Fin inertia	J_L	8.0×10^{-8}	kg m^2
damping	B_0	4.5×10^{-9}	Nm $\frac{s^2}{rad^2}$
Frequency	f_0	4	Hz

$$B_1 \int_{-\pi/2}^{+\pi/2} \Omega_0^2 \cos^2(x)\,dx = B_0 \int_{-\pi/2}^{+\pi/2} \Omega_0^3 \cos^3(x)\,dx$$

and analytically evaluated as:

$$B_1 = B_0 \frac{8}{3\pi} \Omega_0 \tag{6}$$

Therefore, at sinusoidal regime, the damping torque approximated by

$$\widetilde{B}(\tilde{\omega}) = B_0 \frac{8}{3\pi} \Omega_0 \tilde{\omega} \tag{7}$$

with a *peak torque*

$$\left|\widetilde{B}(\tilde{\omega})\right| = B_0 \frac{8}{3\pi} \Omega_0^2 \tag{8}$$

where $|\cdot|$ denotes the amplitude of a sinusoidal variable.

Consequently, the power dissipated against drag is

$$\widetilde{P}_{drag} = \widetilde{B}(\tilde{\omega})\tilde{\omega} = B_0 \frac{8}{3\pi} \Omega_0 \tilde{\omega}^2 \tag{9}$$

with a *peak* power (equivalent to the power amplitude)

$$P_{drag} = \left|\widetilde{P}_{drag}\right| = B_0 \frac{8}{3\pi} \Omega_0^3 \tag{10}$$

4. DC motor selection

4.1. Permanent DC motor model

The electro-mechanical model of a gearless DCM [7], considering a steady-state[2] driving voltage V and current I, can be summarized in the following equations:

$$\begin{cases} V = R_0 I + K_a \omega_m \\ K_a I - \tau_m = J_m \alpha_m + b_r \omega_m \end{cases} \tag{11}$$

where ω_m and α_m are, respectively, the angular speed and the angular acceleration of the rotor, R_0 is the electrical resistance of the armature, K_a is armature constant, J_m is the moment of inertia of the rotor and b_r is the damping constant due to the internal friction (motor bearings), τ_m represents a generic external torque exerted directly on the shaft of the rotor.

4.2. Gearhead characteristics

The gearhead, as other forms of transmission, is an element used to optimally match the motor characteristics with a given load. The gearhead is often treated as an ideal gear ratio N_g. Although this might be acceptable in many cases, especially for high gear ratio it might lead to underestimates of the required power.

In this paper, together with the gear ratio N_g also the gear efficiency η_g will be considered, leading to the following relations between torque τ_m and speed ω_m at the motor and torque τ and speed ω at the load:

$$\omega_m = N_g \omega \tag{12}$$

$$\tau = \eta_g N_g \tau_m \tag{13}$$

in the gearless condition, i.e. when the motor directly drives the load, then such parameters can be conveniently set to $N_g = 1$ and $\eta_g = 1$.

[2] This is equivalent at neglecting the effect of the armature inductance, justified by the fact that electrical dynamics are much faster than mechanical dynamics.

4.3. Overall system equations

A generic resonant load with inertia J_L, stiffness K_L and nonlinear damping as in (1) is described by

$$J_L \alpha + B_0 \omega^2 \mathrm{sign}(\omega) + K_L \theta = \tau \tag{14}$$

where τ as in (13), is the driving torque at gearhead output. By combining Eqs. (11)–(14), the overall system can be described as

$$\begin{cases} V = R_0 I + K_a N_g \omega \\ \eta_g N_g K_a I = \left(\eta_g N_g^2 J_m + J_L\right)\alpha + \left(\eta_g N_g^2 b_m + B_0 \omega \mathrm{sign}(\omega)\right)\omega + K_L \theta \end{cases} \tag{15}$$

4.4. Overall system equations at sinusoidal regime

If the stiffness K_L is selected to ensure that the whole systems resonates at frequency f_0, i.e.

$$K_L = (2\pi f_0)^2 \left(\eta_g N_g^2 J_m + J_L\right) \tag{16}$$

then at sinusoidal regime, using (2)–(4), the inertial torque and the elastic torque will balance one another

$$\left(\eta_g N_g^2 J_m + J_L\right)\tilde{\alpha} + K_L \tilde{\theta} = 0 \tag{17}$$

reducing the final set of equations to:

$$\begin{cases} \widetilde{V} = R_0 \widetilde{I} + K_a N_g \tilde{\omega} \\ \eta_g N_g K_a \widetilde{I} = \left(b_0 + \frac{8}{3\pi} B_0 \Omega_0\right)\tilde{\omega} \end{cases} \tag{18}$$

where $b_0 = \eta_g N_g^2 b_m$. Considering a sinusoidal input voltage $\widetilde{V} = V_0 \cos(2\pi f_0 t)$, armature current $\widetilde{I} = I_0 \cos(2\pi f_0 t)$ and angular velocity as in (3), after simplifying the $\cos(2\pi f_0 t)$ dependence, voltage amplitude V_0 and current amplitude I_0 can be expressed as a function of Ω_0:

$$\begin{cases} V_0 = \frac{R_0}{\eta_g N_g K_a}\left(b_0 + \frac{8}{3\pi} B_0 \Omega_0\right)\Omega_0 + K_a N_g \Omega_0 \\ I_0 = \frac{1}{\eta_g N_g K_a}\left(b_0 + \frac{8}{3\pi} B_0 \Omega_0\right)\Omega_0 \end{cases} \tag{19}$$

which guarantees a one-to-one relationship for all $V_0, I_0, \Omega_0 \geqslant 0$ (as amplitudes are non-negative by definition).

4.5. Power estimates at sinusoidal regime

An instantaneous power balance can be obtained from (18) as

$$\widetilde{V}\widetilde{I} = R_0 \widetilde{I}^2 + \frac{1}{\eta_g}\left(1 + \frac{b_0}{\frac{8}{3\pi} B_0 \Omega_0}\right)\frac{8}{3\pi} B_0 \Omega_0 \tilde{\omega}^2$$

which can be rewritten to highlight the power dissipated against mechanical damping (input electrical power minus electrical losses):

$$\widetilde{P}_{mech} \stackrel{def}{=} \widetilde{V}\widetilde{I} - R_0 \widetilde{I}^2 = \frac{1}{\eta_g}\frac{1}{\eta_x}\widetilde{P}_{drag} \tag{20}$$

where $\widetilde{P}_{drag}$ is as in (9) and η_x is defined as

$$\eta_x = \left(1 + \frac{b_0}{\frac{8}{3\pi} B_0 \Omega_0}\right)^{-1}$$

to take into account the mechanical power dissipated against shaft friction instead of fluid dynamical drag. Therefore, the mechanical power $\widetilde{P}_{mech}$ is dissipated partly in the gearhead (accounted for by η_g) and partly in the shaft friction (accounted for by η_x) before being dissipated against drag.

While efficiencies η_g and η_x can be very close to 1, there is an intrinsic limit to the mechanical power that can be delivered in relation to the electrical losses. The maximum power transfer theorem for linear networks states that, for a given nominal input voltage, the maximum mechanical power delivered to the load cannot exceed the electrical losses, leading to a maximum 50% efficiency which can only be achieved in the case of *impedance-matching* condition. A similar result can be derived here. Substituting the second equation into the first equation of (18) leads to:

$$\widetilde{V} = R_0\widetilde{I} + R_{mech}\widetilde{I}$$

where

$$R_{mech} \stackrel{def}{=} \eta_g\eta_x N_g^2 \frac{K_a^2}{\frac{8}{3\pi}B_0\Omega_0} \tag{21}$$

is the *equivalent electrical impedance* of the mechanical damping, comprising both the drag and the friction at the shaft and gears, as seen at the armature.

The instantaneous mechanical power is therefore

$$P_{mech} = \widetilde{V}\widetilde{I} - R_0\widetilde{I}^2 = \frac{R_0R_{mech}}{(R_0+R_{mech})^2}\frac{\widetilde{V}^2}{R_0} = \mu\frac{1}{2}\frac{\widetilde{V}^2}{2R_0}$$

where

$$\mu \stackrel{def}{=} 4\frac{R_0R_{mech}}{(R_0+R_{mech})^2} = 4\frac{R_{mech}/R_0}{(1+R_{mech}/R_0)^2} \tag{22}$$

is the *impedance-mismatch* factor. Note that $0 < \mu \leqslant 1\ \forall\ R_0, R_{mech} > 0$ and also that $\mu = 1$ if and only if $R_0 = R_{mech}$, meaning that the power dissipated across R_{mech} equals the power dissipated across R_0, i.e. a 50% efficiency. In the best case scenario (i.e. impedance-matching condition: $R_{mech} = R_0$) the total input power is $\widetilde{V}^2/(2R_0)$ and only half of it can be transferred to the mechanical load.

Eq. (20) can now be conveniently rearranged as

$$\eta_g\eta_x\frac{1}{2}\frac{\widetilde{V}^2}{2R_0} = \frac{\widetilde{P}_{drag}}{\mu}$$

or, considering that $\widetilde{V} = V_0\cos(2\pi f_0 t)$ and $\widetilde{P}_{drag} = P_{drag}\cos(2\pi f_0 t)$, in terms of amplitudes

$$\eta_g\eta_x\ \frac{1}{2}\frac{V_0^2}{2R_0} = \frac{P_{drag}}{\mu} \tag{23}$$

where P_{drag} is given in (10).

It should first be noted that (23) is equivalent to the first of (19) as it defines a unique input voltage amplitude V_0 (non-negative by definition) for any desired angular velocity amplitude Ω_0 (also non-negative by definition). The advantage is that (23) provides an interpretation in terms of power and leads to a graphical representation useful for motor selection, as we shall see next.

For a given desired kinematics, i.e. a desired stroke-angle θ_0 at fixed frequency f_0 which uniquely defines Ω_0 from (5), the power P_{drag} to be dissipated against drag is the minimum amount that the motor should be able to handle. In fact, the power *required* from the motor can be even larger in case of impedance-mismatch $\mu < 1$, as indicated by the right-hand side of (23). The left-hand side of (23) represents the *maximum available power* for a given fixed input voltage amplitude V_0. Recalling that the maximum transferrable power to the load is one half of $V_0^2/(2R_0)$, the available power is further reduced by the gear inefficiencies η_g and by the shaft friction η_x. It is therefore advantageous to select motors with optimal impedance match (R_{mech} as close as possible to R_0) so to minimize the required power and not to oversize the motor itself.

Impedance-ratio is also the key to an effective graphical representation. The required power, i.e. the right-hand side of (23), can be plotted in a power vs. impedance-ratio plot independently of any motor–gear selection, and is represented by a family of curves (which we shall refer to as 'drag curves') parameterized by P_{drag}, which in turn only depends on Ω_0. The shape of such curves is solely dictated by μ^{-1}, function of the impedance-ratio R_{mech}/R_0 as defined in (22), and all assume minimum value P_{drag} when $R_{mech}/R_0 = 1$. For example, in the sketch in Fig. 1 two drag curves (dashed line) are drawn corresponding to drags P'_{drag} and P''_{drag} which are relative, respectively, to two different angular speeds Ω'_0 and Ω''_0.

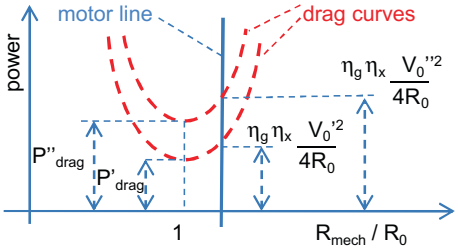


Fig. 1. Power vs. impedance-ratio plot in the *linear* case, i.e. when the impedance-ratio R_{mech}/R_0 remains constant.

As for the left-hand side of (23), let us consider for just a moment the case of linear damping, meaning that the once a motor–gear combination is selected then the R_{mech}/R_0 ratio remains constant. For this linear case, in a power vs. impedance-ratio plot, the left-hand side of (23) would simply be a vertical line (hereafter referred to as 'motor line') parameterized with the input voltage amplitude V_0. In Fig. 1 the vertical solid line represents the locus of points with a fixed impedance-ratio (greater than one, in this specific example) and power proportional to V_0^2. The two drag curves relative to Ω'_0 and Ω''_0 intersect the vertical line in two points which are relative to input voltage amplitudes V'_0 and V''_0. This graphical correspondence defines a one-to-one relationship between V_0 and Ω_0.

Of course, we are interested in nonlinear damping such as (1) in which case the impedance-ratio does depend on the operating speed Ω_0. The left-hand term in (23) can still be represented in a power vs. impedance-ratio plot, the only difference is that it will no longer correspond to a vertical line. In fact, as we shall see in the numerical study in next section, when represented in logarithmic scale, each motor–gear selection can be practically considered as a very steep line[3], especially when compared with the drag curves. Different motor–gear selections can be represented in the same power vs. impedance-ratio plot providing an immediate means of comparison for selecting the best option.

5. Numerical study

In this section, an optimal motor–gearhead combination shall be selected for driving at resonance the wing model presented in Section 2. Among various possibilities of commercial motors and gearheads, the miniature devices from Faulhaber [19] were chosen because of the availability of electro-mechanical parameters for simulations. In particular the motors listed in Table 2 were considered in combination with the gearheads listed in Table 3. As for a matter of nomenclature, the motors in Table 2 will be indicated with numbers 1,...,5 (relative to their ID) while letters 'b',...,'g' shall be used to refer to a particular gear ratio, the letter 'a' is reserved to indicate a gearless combination, in which case an ideal gear ratio $N_{gear} = 1$ and efficiency $\eta = 100\%$ will be assumed. Note

[3] It is possible to roughly estimate theslope in a power vs. impedance-ratio of such lines as follows. Power $P \propto V_0^2$ from the left-hand of (23) and $V_0 \propto \Omega_0^2$, at least for large Ω_0, from (19). Also $R_{mech}/R_0 \propto \Omega_0^{-1}$ from (21). Considering a logarithmic scale power vs. impedance-ratio plot, $\log P \propto 2\log V_0 \propto 4\log\Omega_0$ while $\log(R_{mech/R_0}) \propto -\log\Omega_0$, therefore $\log P/\log(R_{mech}/R_0) \propto -4$.

Table 2
Electro-mechanical parameters relative to five motors from [19]. V_n is the nominal voltage, R_0 is the armature resistance, PWR is the maximum rated power, n_0 is the no-load speed, I_0 is the no-load current, T_{stall} is the torque at stall, $T_{friction}$ is the friction relative to the no-load speed n_0, K_a is the armature constant, J_m is the rotor inertia while n_{max}, T_{max} and I_{max} are the maximum ratings for, respectively, speed, torque and current.

ID	1	2	3	4	5	Units
Series	0615	0816	1016	1219	1024	–
Model	1,5S	003S	003G	012G	012S	–
V_n	1.5	3	3	12	12	V
R_0	3.9	11.5	8.7	69	31.6	Ohm
PWR	0.12	0.17	0.24	0.5	1.11	W
n_0	19,100	15,700	14,200	16,000	14,700	rpm
I_0	0.03	0.016	0.015	0.004	0.004	A
T_{stall}	0.24	0.41	0.64	1.19	2.89	m Nm
$T_{friction}$	0.02	0.04	0.03	0.03	0.03	m Nm
K_a	0.69	1.7	1.93	7	7.7	m Nm/A
J_m	0.01	0.03	0.04	0.13	0.12	g cm^2
Weight	3.5	3.5	6.5	11	8.8	g
n_{max}	13,000	13,000	13,000	12,000	12,000	rpm
T_{max}	0.11	0.15	0.48	0.6	1.28	m Nm
I_{max}	0.341	0.211	0.26	0.1	0.17	A
Gearheads	06/1	08/1	10/1	10/1	10/1	–

Table 3
Mechanical parameters relative to three sets of gearheads to be used in combination with motors from [19].

	b	c	d	e	f	g	
06/1							
N_g	4	16	64	256	1024	4096	
η_g	90	80	70	60	55	48	%
T_g^{max}	25	25	25	25	25	25	m Nm
Weight	2.0	2.8	3.4	4.0	4.4	5.0	g
08/1							
N_g	4	16	64	256	1024	4096	
η_g	90	80	70	60	55	48	%
T_g^{max}	60	60	60	60	60	60	m Nm
Weight	2.9	3.8	4.6	5.4	6.3	7.1	g
10/1							
N_g	4	16	64	256	1024	4096	
η_g	90	80	70	60	55	48	%
T_g^{max}	5	15	54	100	100	100	m Nm
Weight	6.0	7.0	8.0	10	11	13	g

that the last row of Table 2 indicates the compatible gearhead series (i.e. '06/1', '08/1' and '10/1') for each motor.

All parameters required in (18), (19) and (23) can be directly derived from the Tables 2 and 3 with exception of the friction coefficient b_m which can be estimated as $b_m = T_{friction}/n_0$ for each motor.

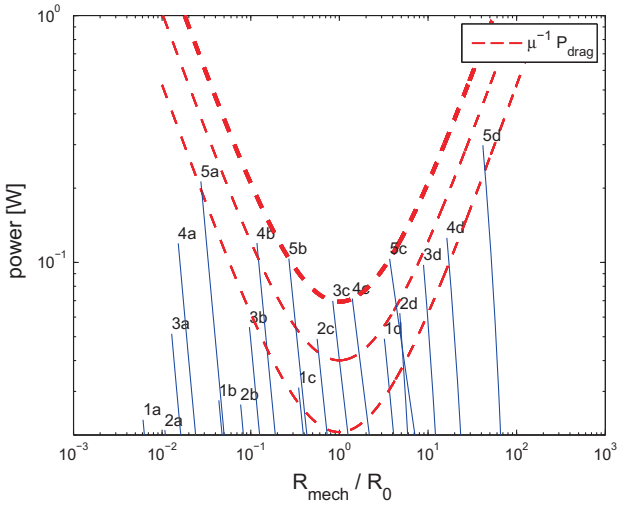


Fig. 2. Power vs. impedance-ratio plot.

Fig. 2 shows the power vs. impedance-ratio plot for all the motors in Table 2 combined with the relative gearheads in Table 3 and for the power drag requirements as in Section 2. The drag curves were drawn for three values of Ω_0 corresponding to 40° (bottom drag curve), 50° (middle drag curve) and 60° (top drag curve) stroke-angle amplitudes (θ_0). The maximum angular speed amplitude Ω_0^{max} is relative to a 60° ($\pi/3$) stroke-angle and equals

$$\Omega_0^{max} = 2\pi f_0 \frac{\pi}{3} \approx 26.3 \text{ rad/s}$$

For each motor–gearhead combination, 60 equally spaced (each corresponding to 1° increment in stroke-angle) possible operating angular speed amplitudes $\Omega_0^i \in [0\ \Omega_0^{max}]$ were considered and for each operating speed the required input voltage V_0^i and current I_0^i amplitudes were computed via (19).

Although solutions of (19), large values of input voltage V_0^i, current I_0^i, input power $V_0^i I_0^i$ or speed Ω_0^i will likely exceed some intrinsic limit of the motor–gearhead combination. For this reasons, for each motor–gearhead combination, only those operating conditions would be represented in the power vs. impedance-ratio plot which satisfy at the same time *all* the constraints as given in Table 4. Fig. 2 represents for each motor–gearhead combination such admissible operating conditions. As mentioned in previous section, in a logarithmic scale the left-hand side of (23) appears as a steep (almost vertical) curve for each motor–gearhead combination.

Fig. 2 allows to compare *at once* the performance of all the available motor–gearhead combinations. It can be appreciated that not all the motor lines reach the 60° drag curve. For example, motors 1 and 2 (which are the smallest in weight, see Table 2) with all the possible gearhead combinations (a–g) have not enough power to reach the 60° drag curve. The combination 2c (motor 2 with a 16 gear ratio gearhead) shows the most favorable impedance-ratio

Table 4
Constraints.

Limit	Condition	
Nominal voltage	V_0^i	$\leqslant V_n$
Max current	I_0^i	$\leqslant I_{max}$
Max power	$V_0^i I_0^i$	$\leqslant$PWR
Max motor speed	$\Omega_0^i N_g$	$\leqslant n_{max}$
Max motor torque	$\frac{8}{3\pi} B_0 {\Omega_0^i}^2 / N_g$	$\leqslant T_{max}$
Max gear torque	$\frac{8}{3\pi} B_0 {\Omega_0^i}^2$	$\leqslant T_g^{max}$

(i.e. the closest to 1) and is the only combination involving motor 2 that can provide more than 50° stroke-angle.

In fact the only combinations that can reach the 60° drag curve are the 3c, 4c, 5b and 5c. Due to large impedance-mismatch, combination 5b and 5c can indeed produce the desired output Ω_0^{max} but at a cost of a larger required power (≈100 mW, note that the motor 5 is also the most powerful of the series). On the other hand, combinations 3c and 4c are equally able to produce the desired kinematic output Ω_0^{max} with lower effort (≈70 mW) as the impedance-mismatch factor is close to 1. Of the two, combination 3c is to be preferred because of a smaller weight (6.5 g for motor 3 vs. 11 g for motor 4, for both combinations the gearhead c would add 7 g).

Remark. It is worth noting that, since the slopes of the motor lines are negative, motor lines that start with an impedance-ratio larger than 1 for small output speed, gain a better impedance-ratio (i.e. closer to 1) as speed increases. The impedance-ratio counts more at higher speed which is when most of the power is required.

6. Accuracy of the sinusoidal regime hypothesis

The Eqs. (18)–(23) are based on the sinusoidal regime hypothesis. Here numerical simulations are presented to assess the level of accuracy of such an assumption. In particular, the nonlinear ordinary differential equations (ODE) in (15) is numerically solved without the sinusoidal regime simplifying hypothesis. All numerical parameters required for the simulations can be found in Section 2 as well as in Tables 2 and 3, the stiffness K_L which ensures resonance at $f_0 = 4$ Hz is determined as in (16).

To determine how indicative the proposed impedance-matching based method can be, motors 3, 4 and 5 (which proved powerful enough according to the proposed method to produce the desired output kinematics) were numerically tested with a variety of gear ratios. In particular, we simulated the motor performance with any integer gear ratio between 1 and 100, considering an ideal 100% efficiency for all ratios except for the actually available 4, 16, and 64 ratios, for which the efficiencies as in Table 3 were used instead.

Similarly to the framework of the maximum power transfer theorem, which holds for linear networks and which was here proved under the sinusoidal regime assumption, we also assumed a constant amplitude voltage $V = V_0\cos(2\pi f_0 t)$ as an input to (15). The system was implemented in MATLAB (MathWorks Inc.) and the ODE was solved with the built-in `ode45()` function. Since we were only interested in the steady-state solutions, we started with arbitrary initial conditions $\theta(0) = 0$ and $\omega(0) = 0$ and simulated the system for exactly one period of time $T = 1/f_0$. Then the simulation was repeated using the final values $\theta(T)$ and $\omega(T)$ as initial conditions for the new simulation. This step was repeated until the relative error between the current solutions and the solutions provided by the previous simulation step dropped below 0.1% (maximum error throughout the period).

Fig. 3 shows the response for motors 3, 4, and 5. As the motors are different, also different fixed input voltage amplitudes were used (2 V, 5 V and 4 V, respectively, for motor 3, 4, and 5), heuristically chosen to provide an approximately 60° output stroke-angle for the 3c, 4c, 5b and 5c cases. The vertical lines in Fig. 3 indicate the actually available gear ratios 4, 16, 64 as well as the gearless case $N_g = 1$.

The situation depicted in Fig. 2 is confirmed by such numerical simulations. For each plot in Fig. 3, it can be recognized what the optimal gear ratio would be for each case (e.g. for motor 5, $N_g = 10$ would give the maximum Ω_0 output). Although the actually available gear ratios never perfectly match the optimal ones,

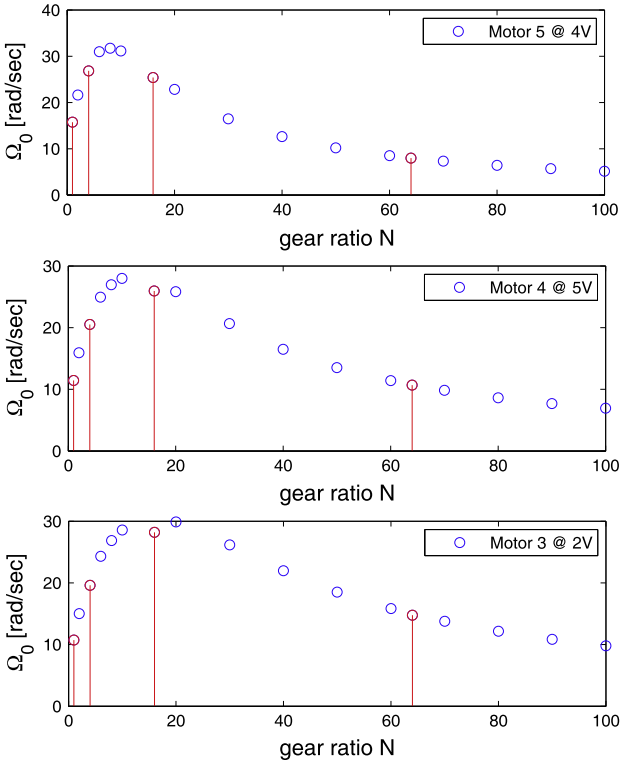


Fig. 3. Steady-state kinematics, in terms of angular speed amplitude Ω_0, vs. gear ratios for motors 3 (bottom plot), 4 (middle plot) and 5 (top plot). Only the gear ratios 1, 4, 16, 64 (indicated by vertical lines) represent available motor gearhead combinations, for all the other gear ratios an ideal 100% efficiency was assumed.

it can be recognized that 3c and 4c (corresponding to the third vertical line, respectively, in the bottom and middle plot of Fig. 3) are very close to optimality. Also note that since for the available gear ratios the actual non-ideal efficiency was used, their output is slightly smaller than expected from the whole trend of ideal gears.

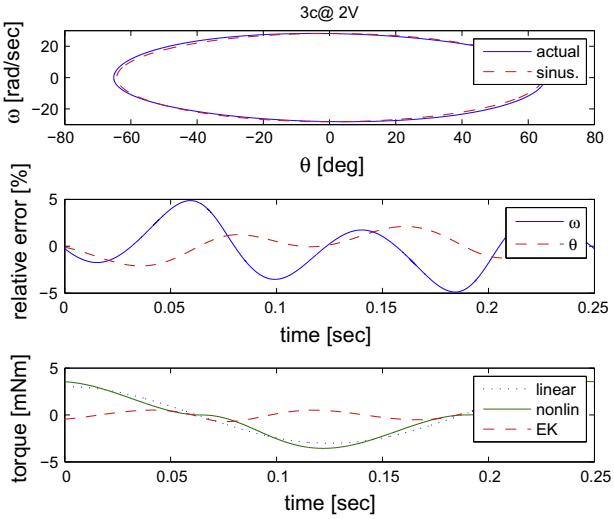


Fig. 4. Steady-state output relative to the motor–gearhead combination '3c' (motor 3 with a 16 gear ratio). Top plot: phase-space ω vs. θ of the steady-state kinematics (sold line) superimposed with sinusoidal regime assumption solution (dashed line). Middle plot: relative error between the steady-state solution and the sinusoidal regime solution vs. time, for angular velocity (solid line) and angular position (dashed line). Bottom plot: superposition of nonlinear torque (solid line), linear equivalent torque (dotted line) and elasto-kinetic torque (dashed line) vs. time.

Finally, for the 3c combination which from the previous section turned out to be the most suitable, the actual output is reported. Fig. 4 shows the phase-space ω vs. θ plot (top graph) of the actual steady-state solutions of (15) superimposed with pure sinusoidal solutions assumed under the sinusoidal regime hypothesis. As expected there is very little difference. In particular, relative errors between the actual and the pure sinusoidal solutions is within 5% for the ω and within 3% for θ (middle plot in Fig. 4). The output torques are also reported in the bottom plot of Fig. 4. In particular, the nonlinear torque as in (1) superimposed with the linear approximant as in (7) are presented as well as the elasto-kinetic (EK) contribution. The latter is in fact the torque which is *not* compensated for by the resonant system. The actual steady-state solutions of (15) are not in fact purely sinusoidal, although periodic, meaning that higher harmonics are also present. The mechanical resonance is capable of perfectly balancing the elastic and inertial (i.e. kinetic) torques only for the first harmonic (at frequency f_0). For higher harmonics (i.e. frequencies nf_0, with $n > 1$) the inertial torque will be larger than the elastic one, in particular

$$-(2\pi n f_0)^2\left(\eta_g N_g^2 J_m + J_L\right) + K_L \neq 0, \quad \forall\, n > 1 \tag{24}$$

It should be noted that since $B(\omega)$ in (1) is a *odd* nonlinearity, i.e. $B(-\omega) = -B(\omega)$, the steady-state periodic solutions will only contain odd frequencies. There is no contribution for the second harmonic and the first higher harmonic is in fact the third. This can also be appreciated in the bottom plot of Fig. 4 where the elasto-kinetic (EK) torque is clearly periodic of period $1/(3f_0)$.

7. Conclusion

This paper presents a novel method for selecting the most appropriate motor–gearhead combination for driving nonlinearly damped loads at resonance. This loading condition is common in biomimetic robotics, when designing propulsion systems based on flapping appendages, such as fins or wings, in a viscous fluid such as water or air. Specifically to this condition, a typically nonlinear, quadratic damping is considered. The power estimates derived in this paper allow representing the problem on a power vs. impedance-mismatch plot, as in Fig. 2, where all available motor–gearhead combinations can be compared at once with respect to the output power requirements.

To derive power estimates, we first determine the dynamic equations for the full system (15) taking into account: (i) the electro-mechanical equations of the DC motor; (ii) the gearhead characteristics (in particular the transmission efficiency); (iii) the load, which consists of the inertia and nonlinear damping of the wing/fin as well as of the elastic properties of the coupling mechanism.

Such a loading condition reflects an important biomimetic aspect: *resonance*. It is in fact biological evidence that the efficiency behind locomotion based on wings/fins-flapping is largely due to the elastic properties of muscles, or to the external cuticle in the case of insects, which induce structural resonance when coupled with the inertia of the flapping appendages themselves. The presence of such elastic components for storage and release of kinetic energy avoids in fact unnecessary waste of inertial power (required to accelerate/decelerate wings or fins).

The theory of linear networks suggests that, to maximize power transmission when driving a resonant load, the optimal gear ratio should be chosen so that the resistive impedance of the load, reflected at the actuator, matches the impedance of the actuator itself. The proposed method extends the linear case to the nonlinear one based on a second biological observation: the wings/fins kinematics observed in nature can be considered *harmonic* with a good approximation.

Considering steady-state solutions which are approximately sinusoidal allows us to remarkably simplify analytical calculations and derive approximate solutions which closely match the numerical solutions of the original system, e.g. see Fig. 4 (top). In particular, numerical steady-state solutions derived from the full nonlinear dynamics (15) provide support for the simplifying sinusoidal regime assumption, as the difference between the numerical steady-state solutions and pure sinusoids is lower than 5% and 3% for the angular speed and position, respectively, for the optimal motor–gearhead combination.

The fact that such a good matching also holds at large strokes (±60°) can be explained with two main observations: (i) we are not over-simplifying the system via some mere linearization but we are maintaining its central feature, i.e. the nonlinear damping; (ii) for the specific loading condition of interest (14) the nonlinearity is *only* in the damping and not in the elastic or inertial terms.

The latter observation has profound practical implications since, in order to fully exploit resonance, an elastic system should be designed to perfectly[4] couple with the inertia to satisfy Eq. (16) and this is conveniently done when both inertial and elastic terms are *linear*. For the inertial term to maintain linearity at large strokes, direct driving of the load is advised. As also underlined by Baek, Ma and Fearing [2], commonly employed transmissions such as slider–crank mechanisms suffer from fundamental geometric nonlinearities, directly reflected in the inertial term. Such systems, when coupled with springs for energy recovery, fail at fully exploiting resonance due to the inherent kinematic nonlinearity.

Linearity requirements for the elastic term at large deformations call for a careful selection of both materials and structures. It is not uncommon, for large stroke requirements, that nonlinear elastic behaviors heavily affect the resonance and therefore the method proposed in this paper, e.g. previous work [24,4] on flapping wings reported and modeled resonant frequency shifts due to the so-called 'softening' of piezoelectric actuators at higher fields. Although technologically challenging, achieving linearity of elastic structures over large range of deformations and frequencies is indeed feasible, as demonstrated by the various examples of *resonant optical scanners*[5] currently used in laser applications [20]. To the author's knowledge, this is the only application where magnetic torque motors (with working principles similar to regular permanent magnet DC motors) are used to *directly* drive a resonant load, typically a mirror used to reflect a laser beam for optical scanning purposes. Mechanical resonance between the elastic structure (typically a torsional rod or a flexure joint [20]) and the inertia of the mirror is used to minimize the driving torque required from the motor. Nevertheless, an oscillating mirror (often resonating in vacuum) is different from a flapping wing for which nonlinear aerodynamic damping must be considered but, despite such important differences, resonant scanners demonstrate that it is indeed possible to efficiently[6] and directly drive a resonant load at large angular displacements.

In conclusion, this paper provides a method for selecting DC motors for driving nonlinearly damped, resonant loads based on 'quick' power estimates inspired by Weis-Fogh's analysis [29], whose quotation is in order here:

[4] It should be noted that such a 'perfect' coupling only holds for the first harmonic, i.e. for Eq. (17), where elastic and inertial term perfectly cancel each other at steady-state (a well known fact from second-order linear systems). This balance does not hold for higher harmonics, leading to the inequality (24). The inertial and elastic terms do have an effect for higher harmonics, resulting in a non-zero elasto-kinetic torque still required from the motor, as shown in Fig. 4 (bottom plot).

[5] Also known as resonant galvanometers or 'galvos'.

[6] The efficiency resulting primarily in very compact active devices, see for example the SC-3 sub-miniature model from Electro-Optical Products Corporation (http://www.eopc.com/), where a 6 mm diameter, 1 mm thick aluminum mirror oscillates at 250 Hz with a 50° angular stroke.

"*Although entitled 'Quick estimates', this does not mean that the approach is superficial, but rather that a procedure has been devised whereby the flight performance of a given animal can be evaluated quantitatively as well as qualitatively on the basis of only a few accurate observational data and a minimum of computation.*"

– *verbatim* from [29, p. 169]

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References

[1] Ayers J, Davis JL, Rudolph A. Neurotechnology for biomimetic robots. MIT Press; 2002.
[2] Baek SS, Ma KY, Fearing RS. Efficient resonant drive of flapping-wing robots. In: Proceedings of the IEEE/RSJ international conference on intelligent robots and systems (IROS), October 11–15, St. Louis, MO, USA; 2009.
[3] Blake RW. The energetics of hovering flight in the mandarin fish (*Synchropus picturatus*). J Exp Biol 1979;82:25–33.
[4] Campolo D. Energy efficient driving of piezoelectric actuators for the micromechanical flying insect. PhD Thesis, Scuola Superiore Sant'Anna in Pisa, Italy; 2001.
[5] Colgate JE, Lynch KM. Mechanics and control of swimming: a review. IEEE J Oceanic Eng 2004;29:660–73.
[6] Cox A, Monopoli D, Cveticanin D, Goldfarb M, Garcia E. The development of elastodynamic components for piezoelectrically actuated flapping micro-air vehicles. J Intell Mater Syst Struct 2002;13:611–5.
[7] Craig JJ. Introduction to robotics: mechanics and control. 2nd ed. Prentice Hall; 1989.
[8] Cusimano G. Optimization of the choice of the system electric drive-device-transmission for mechatronic applications. Mech Mach Theory 2007;42: 48–65.
[9] Cusimano G. Generalization of a method for the selection of drive systems and transmissions under dynamic loads. Mech Mach Theory 2005;40:530–58.
[10] Dickinson MH, Lehmann FO, Sane SP. Wing rotation and the aerodynamic basis of insect flight. Science 1999;284:1954–60.
[11] Gelb A, Vander Velde WE. Multiple-input describing functions and nonlinear systems design. New York: Mac Grow-Hill; 1968.
[12] Hove JR, O'Bryan LM, Gordon MS, Webb PW, Weihs D. Boxfishes (Teleostei: Ostraciidae) as a model system for fishes swimming with many fins: kinematics. J Exp Biol 2001;204:1459–71.
[13] Kawamura Y, Souda S, Nishimotor S, Ellington C. Clapping-wing micro air vehicle of insect size. In: Kato N, Kamimura S, editors. Biomechanisms of swimming and flying: fluid dynamics, biomimetic robot, and sports science. Springer; 2008. p. 319–30.
[14] Kodati P, Hinkle J, Deng X. Micro autonomous robotic ostraciiform (MARCO): design and fabrication. In: Proceedings of the IEEE international conference on robotics and automation (ICRA), Roma, Italy; 10–14 April, 2007. p. 960–5.
[15] Lachat D, Crespi A, Ijspeert AJ. BoxyBot: a swimming and crawling fish robot controlled by a central pattern generator. In: Proceedings of the first IEEE/RAS-EMBS international conference on biomedical robotics and biomechatronics (BIOROB), Pisa, Italy; 20–22 February, 2006. p. 643–8.
[16] Licht S, Polidoro V, Flores M, Hover FS, Triantafyllou MS. Design and projected performance of a flapping foil AUV. IEEE J Oceanic Eng 2004;29:786–94.
[17] Lindstedt SL, Reich TE, Keim P, La Stayo PC. Do muscles function as adaptable locomotor springs? J Exp Biol 2002;205:2211–6.
[18] Madangopal R, Khan ZA, Agrawal SK. Biologically inspired design of small flapping wing air vehicles using four-bar mechanisms and quasi-steady aerodynamics. J Mech Des 2005;127:809–16.
[19] http://www.micromo.com/.
[20] Montagu J. Scanners – galvanometric and resonant. In: Drigger RG, editor. Encyclopedia of optical engineering, vol. 3. Marcel Dekker, Inc.; 2003.
[21] Roos F, Johansson H, Wikander J. Optimal selection of motor and gearhead in mechatronic application. Mechatronics 2006;16:63–72.
[22] Sanders S. On limit cycles and the describing function method in periodically switched circuits. IEEE Trans Circ Syst – I: Fund Theory Appl 1993;40:564–72.
[23] Sastry SS. Nonlinear systems: analysis, stability, and control. New York: Springer-Verlag; 1999.
[24] Sitti M, Campolo D, Yan J, Fearing RS. Development of PZT and PZN-PT based unimorph actuators for micromechanical flapping mechanisms. In: IEEE international conference on robotics and automation (ICRA), Seoul, Korea; May 21–26, 2001. p. 3839–46.
[25] Triantafyllou MS, Techet AH, Hove FS. Review of experimental work in biomimetic foils. J Oceanic Eng 2004;29:585–94.
[26] Van de Straete HJ, Degezelle P, de Shutter J, Belmans R. Servo motor selection criterion for mechatronic application. IEEE/ASME Trans Mechatronics 1998;3:43–50.
[27] Van de Straete HJ, de Shutter J, Belmans R. An efficient procedure for checking performance limits in servo drive selection and optimization. IEEE/ASME Trans Mechatronics 1999;4:378–86.
[28] Weis-Fogh T. Energetics of hovering flight in hummingbirds and in *Drosophila*. J Exp Biol 1972;56:79–104.
[29] Weis-Fogh T. Quick estimates of flight fitness in hovering animals, including novel mechanisms for lift production. J Exp Biol 1973;59:169–230.
[30] Yasuda T, Shimoyama I, Miura H. Microrobot actuated by a vibration energy field. Sens Actuators A 1994;43:366–70.